

Indraneel Chatterjee

Machine Learning Engineer Intern | Data Science & AI Systems

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🐙 github.com/DataDetective-Neel | 🌐 Portfolio

Machine Learning engineer with hands-on experience in **end-to-end ML pipelines, NLP systems, and data-driven application development**. Built production-grade systems using **Python, Scikit-learn, HuggingFace Transformers, FastAPI, Docker, and Power BI**. Demonstrated applied ML across **regression, NLP, and LLM-based architectures** through real-world projects and hackathon deployments. Co-authored a **published research contribution** in stochastic modelling; exploring **LLM evaluation, model interpretability, and deep learning** with Keras and TensorFlow.

Education

NIIT University, Neemrana

Aug 2024 – Jul 2028

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.24 / 10

Relevant Coursework: Data Structures, Design & Analysis of Algorithms, Object Oriented Programming, Discrete Mathematics, Probability & Random Processes, Computer Architecture & Organisation, Calculus, Algebra & Differential Equations

KPS Udaan

Completed 2024

Senior Secondary (Non-Medical)

92.8%

Secondary

96.6%

Professional Experience

Sponsorship Core Member | Entente 2026 – International Conference, NIIT University

Dec 2025 – Apr 2026

- Managed corporate partnership strategy and stakeholder relationships for a university-level international conference, coordinating cross-functional outreach and sponsorship pipelines.
- Drove end-to-end sponsorship initiatives from prospecting to closure, applying structured communication and negotiation workflows across multiple concurrent leads.

Sponsorship Lead | siNUsoid – NIIT University Tech Fest

Apr 2025 – Dec 2025

- Led partnerships and funding outreach for the university's flagship annual tech event; coordinated cross-team execution across design, logistics, and PR verticals.
- Secured the highest sponsor count in siNUsoid's recorded history**, demonstrating data-driven outreach targeting and measurable stakeholder conversion.

Learning Assistant – Research & Planning Desk | CoEET, NIIT University

Jan 2025 – Mar 2025

- Assisted research initiatives at the **Centre of Excellence in Educational Technology**, supporting data collection, analysis workflows, and strategic planning activities.
- Applied **structured data interpretation and reporting** to assist institutional decision-making – directly translating to data pipeline and analytical thinking skills relevant to ML roles.

Key Projects

AeroSphinx – PM2.5 Air Quality Prediction

ISRO Hackathon 2025



- Built an end-to-end **Random Forest regression pipeline** integrating **satellite AOD data with ground-sensor readings** for PM2.5 air quality forecasting across temporal-spatial domains.
- Engineered **time-aware feature engineering, spatial alignment, and cross-validation workflows** to improve model generalization on heterogeneous real-world sensor data.
- Evaluated model performance using **MAE, RMSE, and R² metrics**; visualized prediction outputs and feature importances via **Power BI** dashboards for stakeholder communication.

Sentiment Analysis Dashboard

SIH 2025 – College Level Qualified

- Designed a **cloud-based NLP pipeline** using **DistilBERT and DistilBART (HuggingFace Transformers)** for multi-class sentiment classification and abstractive summarization of large comment datasets.
- Deployed a **production-grade backend** using **FastAPI + Celery + Redis + Docker**, supporting async task queues, audit logging, and PDF/Word export workflows at scale.
- Integrated **interactive analytics visualizations** enabling real-time insight delivery – reducing manual review effort through automated NLP-driven categorization.

PipelineIQ – Autonomous CI/CD Failure Resolution System



- Architected a **distributed, event-driven system** using **Kafka-based streaming pipelines** to ingest GitHub Actions workflow events and route failures for automated diagnosis.
- Engineered an **LLM-based log analysis engine** to automatically detect root causes from large-scale build logs, classify change risk, and generate deterministic code fixes via PR automation.

- Reduced mean debugging time by **>90%** through end-to-end automated remediation workflows, replacing manual developer intervention with policy-driven resolution logic.

DocuMentor – AI-Powered Document Assistant



- Built a **fully offline, browser-native AI assistant** using **WebAssembly and WebGPU** for on-device model inference – zero data leaves the user’s device, enforcing strict privacy by design.
- Implemented **three document intelligence modes**: Smart Highlights (priority-ranked AI annotations), Student Mode (interactive quiz & analogy generation), and Research Mode (argument critique & rewriting).
- Engineered a **Privacy Scanner** to detect sensitive artifacts (AWS keys, JWTs) pre-inference, and a **dedicated Web Worker** architecture to maintain UI responsiveness during GPU-accelerated generation.

Publications

Book Chapter | *Foundations and Emerging Trends in Contemporary Multidisciplinary Science*

April 2026

DOI: <https://doi.org/10.5281/zenodo.19692035> Google Scholar

- Co-authored “**Relative Aging Comparisons of Two Coherent Systems under Dependent Identically Distributed Components and System Signatures**” – published by **Cogniverse Press** (ISBN: 978-93-47652-64-6).
- Developed theoretical results using **stochastic ordering, distortion function representations, cumulative hazard rate analysis, and system signature frameworks** to compare aging behavior of coherent reliability systems.
- Applied **probabilistic modelling and mathematical proof techniques** to establish sufficient conditions for relative aging comparisons – with direct applications to **k-out-of-n, series, and parallel system configurations**.

Achievements & Certifications

- **Participant | ISRO Hackathon 2025**: engineered **AeroSphinx**, a end-to-end **PM2.5 air quality forecasting pipeline** integrating satellite AOD and ground-sensor data using **Random Forest regression, temporal-spatial feature engineering, and Power BI visualization** – directly addressing a real-world environmental monitoring challenge. [GitHub](#)
- **College Level Qualified | Smart India Hackathon 2025**: built a **production-grade NLP sentiment analysis system** using **Distil-BERT, FastAPI, Docker, and Redis** – selected among top teams at institutional level for national-stage consideration.
- **Published Book Chapter | Cogniverse Press**: co-authored “*Relative Aging Comparisons of Two Coherent Systems under Dependent Identically Distributed Components and System Signatures*” in *Foundations and Emerging Trends in Multidisciplinary Sciences* – applying **stochastic modelling, probability theory, and system reliability analysis**.
- **IBM Data Science Professional Certificate** | *Coursera (IBM)* [Certificate](#)
Data Science, Machine Learning, Python, SQL, Data Visualization, Model Evaluation
- **Python 3 Programming Specialization** | *Coursera (University of Michigan)* [Certificate](#)
Python, OOP, Data Structures, REST APIs, Pandas, Pillow, OpenCV Basics